6.5 Classwork: Quadratic Functions and Symmetry — Markscheme

1. The diagram shows the graph of a quadratic function with x-intercepts at $(1, 0)$ and $(5, 0)$.
   (a) Find the equation of the axis of symmetry. [1]
   (b) Write the equation of the parabola in factored form, $y = (x - p)(x - q)$. [1]
   (c) Expand to show that the equation is $y = x^2 - 6x + 5$. [2]
   (d) Find the coordinates of the vertex. [2]
   (e) Write down the y-intercept. [1]

Solutions

(a) $x = \dfrac{1+5}{2} = 3$ **(A1)**

The axis of symmetry is $x = 3$.

(b) $y = (x - 1)(x - 5)$ **(A1)**

(c) $y = (x - 1)(x - 5)$

$= x^2 - 5x - x + 5$ **(M1)**

$= x^2 - 6x + 5$ **(AG)**

(d) $y = (3)^2 - 6(3) + 5 = 9 - 18 + 5 = -4$ **(M1)**

Vertex: $(3, -4)$ **(A1)**

(e) y-intercept: $x = 0$, $y = 0 - 0 + 5 = 5$, so $(0, 5)$ **(A1)**

2. The diagram shows the graph of $g(x) = -(x-2)^2 + 9$.

   (a) Write down the coordinates of the vertex. [1]

   (b) Write down the equation of the axis of symmetry. [1]

   (c) Show that $g(x) = -x^2 + 4x + 5$. [2]

   (d) Solve $g(x) = 0$ to find the x-intercepts. [3]

Solutions

(a) Vertex: (2, 9) **(A1)**

(b) $x = 2$ **(A1)**

(c) $g(x) = -(x-2)^2 + 9$

$= -(x^2 - 4x + 4) + 9$ **(M1)**

$= -x^2 + 4x - 4 + 9$

$= -x^2 + 4x + 5$ **(AG)**

(d) $-x^2 + 4x + 5 = 0$

$x^2 - 4x - 5 = 0$ (multiply both sides by -1) **(M1)**

$(x-5)(x+1) = 0$ **(M1)**

$x = 5$ or $x = -1$ **(A1)**

The x-intercepts are $(-1, 0)$ and $(5, 0)$.

3. The diagram shows the graph of $h(x) = x^2 - 2x - 3$.

   (a) Write $h(x)$ in factored form, $(x-p)(x-q)$. [2]

   (b) Write down the x-intercepts of the parabola. [1]

   (c) Find the equation of the axis of symmetry. [1]

   (d) Find the coordinates of the vertex. [2]

   (e) Write $h(x)$ in vertex form, $(x-h)^2 + k$. [1]

Solutions

(a) Find two numbers that multiply to -3 and add to -2: **(M1)**

$h(x) = (x+1)(x-3)$ **(A1)**

(b) x-intercepts: $(-1, 0)$ and $(3, 0)$ **(A1)**

(c) $x = \dfrac{-1+3}{2} = 1$ **(A1)**

(d) $h(1) = (1)^2 - 2(1) - 3 = 1 - 2 - 3 = -4$ **(M1)**

Vertex: $(1, -4)$ **(A1)**

(e) $(x-1)^2 - 4$ **(A1)**

4. Consider the quadratic function $p(x) = -x(x-4)$.
   (a) Write down the x-intercepts. [1]
   (b) Find the equation of the axis of symmetry. [1]
   (c) Find the coordinates of the vertex. [2]
   (d) Expand to write $p(x)$ in the form $ax^2 + bx + c$. [1]
   (e) On the grid below, sketch the graph of $p(x)$. Label the vertex and x-intercepts. [2]

Solutions

(a) x-intercepts: (0, 0) and (4, 0) **(A1)**

(b) $x = \dfrac{0+4}{2} = 2$ **(A1)**

(c) $p(2) = -2(2-4) = -2(-2) = 4$. Vertex: (2, 4) **(M1)(A1)**

(d) $p(x) = -x^2 + 4x$ **(A1)**

(e) Graph: downward-opening parabola through (0, 0) and (4, 0) with vertex at (2, 4).
(A1) shape, **(A1)** key points